

DRAWING LIST FOR TEST PACKAGE

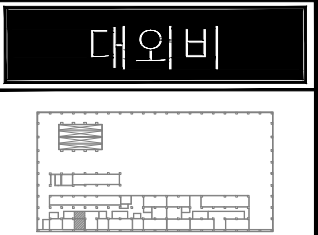
PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

H3PO4-447

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	569	4FA	OD25-H3PO4-HP2-N-7002-01	1 OF 2	0	CJ-M15W13-PR-A20-000059	2	
2	569	4FA	OD25-H3PO4-HP2-N-7002-02	2 OF 2	0	CJ-M15W13-PR-A20-000059	2	
3	569	4FA	OD25-H3PO4-HP2-N-7003-01	1 OF 1	0	CJ-M15W13-PR-A20-000059	2	
4	569	4FA	OD25-H3PO4-HP2-N-7004-01	1 OF 1	0	CJ-M15W13-PR-A20-000059	2	
5	569	3FA	OD25-H3PO4-HP2-N-7002-01	1 OF 1	0	CJ-M15W14-PR-A20-000505	1A	
6	569	2FA	OD25-H3PO4-HP2-N-7002-01	1 OF 1	0	CJ-M15W14-PR-A20-000505	1A	
7	569	2FA	OD25-H3PO4-HP2-E-7002-01	1 OF 1	1	CJ-M15W14-PR-A20-000505	1A	

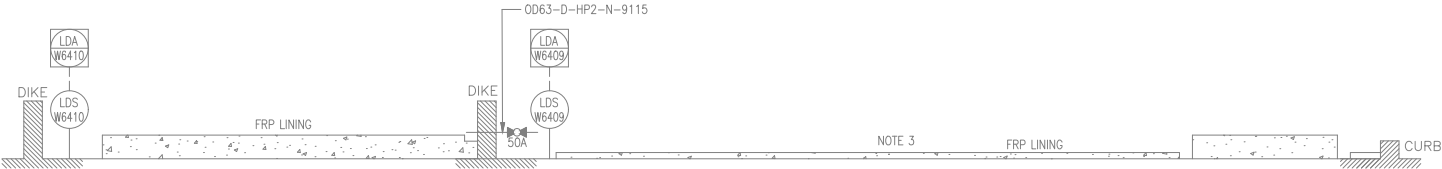
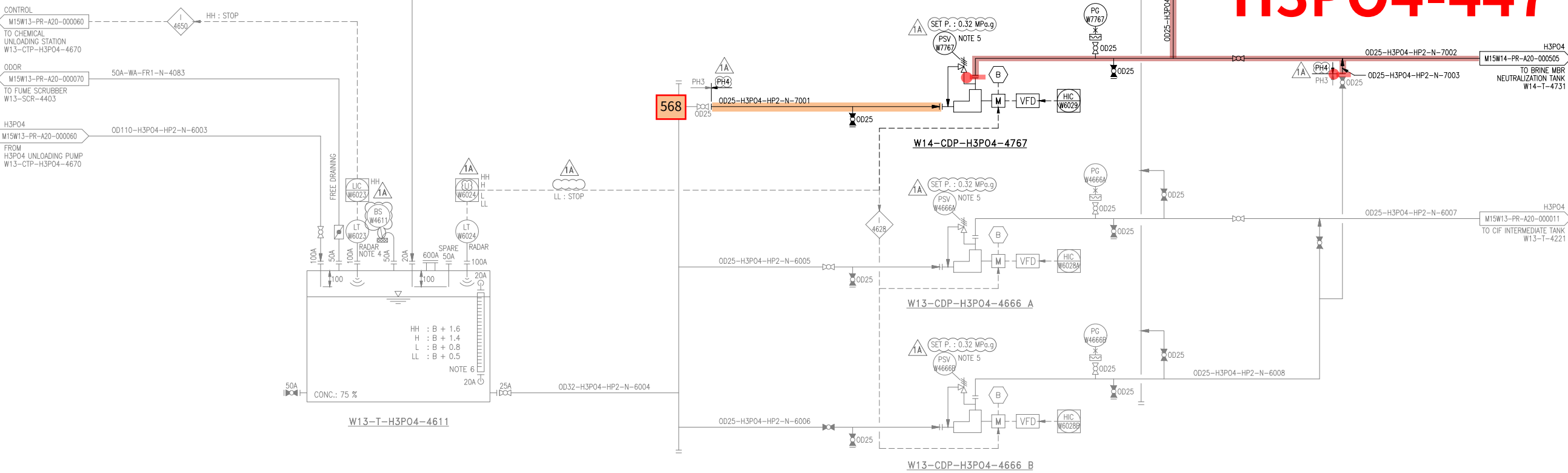
ITEM NO.			W13-T-H3PO4-4611		
SERVICE			H3PO4 STORAGE TANK		
Q'TY			1		
TYPE			VERTICAL CYLINDRICAL		
SIZE	ID x H	m	2.1 x 2.0 (EH 1.6)		
CAPA.	m3		7 (EH 6)		
MAT'L			STS304 + FRP		



KEY PLAN (4F)

- NOTE
1. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM / CHECK BALL
 2. 약품종류에 따라 LINE 색상을 구분하여야 한다.
 3. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 침수되어, DRAIN HEADER로 이동한다.
 4. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
 5. PUMP VENDOR 공급.
 6. TANK의 역면 레벨 게이지는 VENDOR에서 공급한다.
 7. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
 8. INTERLOCK NO. I-4650
(W13-CTP-H3PO4-4670 AUTO STOP)
: REMOTE STATUS에서 W13-T-H3PO4-4611 LEVEL (LI-W6023) HIGH HIGH STOP.
 9. INTERLOCK NO. I-4628
(W13-CDP-H3PO4-4666 A/B AUTO STOP)
: REMOTE STATUS에서 W13-T-H3PO4-4611 LEVEL (LI-W6024) LOW LOW STOP.
 10. FOR REUSE PUMP CONNECTION NOZZLE.
 11. DELETED.
 12. OPERATING CAPACITY : 2 LPM
OPERATING PRESSURE : 0.29 MPa
 13. OPERATING CAPACITY : 0.5 LPM
OPERATING PRESSURE : 0.29 MPa

H3PO4-447



ITEM NO.	W13-CDP-H3PO4-4666 A/B
SERVICE	H3PO4 FEED PUMP
Q'TY	1 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa 1A (MAX. 2.1 x MAX. 1 NOTE 12)
MAT'L	PVDF/PTEF/CERAMIC
POWER kW	0.4 1A

ITEM NO.	W14-CDP-H3PO4-4767
SERVICE	H3PO4 FEED PUMP
Q'TY	1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa 1A (MAX. 0.52 x MAX. 1 NOTE 13)
MAT'L	PVDF/PTEF/CERAMIC
POWER kW	0.4 1A

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31
수정 REV.	설명 DESCRIPTION	날짜 DATE

제 도	이 인 학
DRAWN BY	
검 토	한 진 희
CHECKED BY	
승 인	박 동 영
APPROVED BY	
제 작 사	

SK 에코플랜트	
공 사 명	
PROJECT TITLE	
M15X Ph4 WWT Project	
PROJECT NO.	
22330D	

2	REVISED AFC	24.06.21
1	REVISED AFC	23.02.28
0	APPROVAL FOR CONSTRUCTION	22.09.23
수정 REV.	설명 DESCRIPTION	날짜 DATE

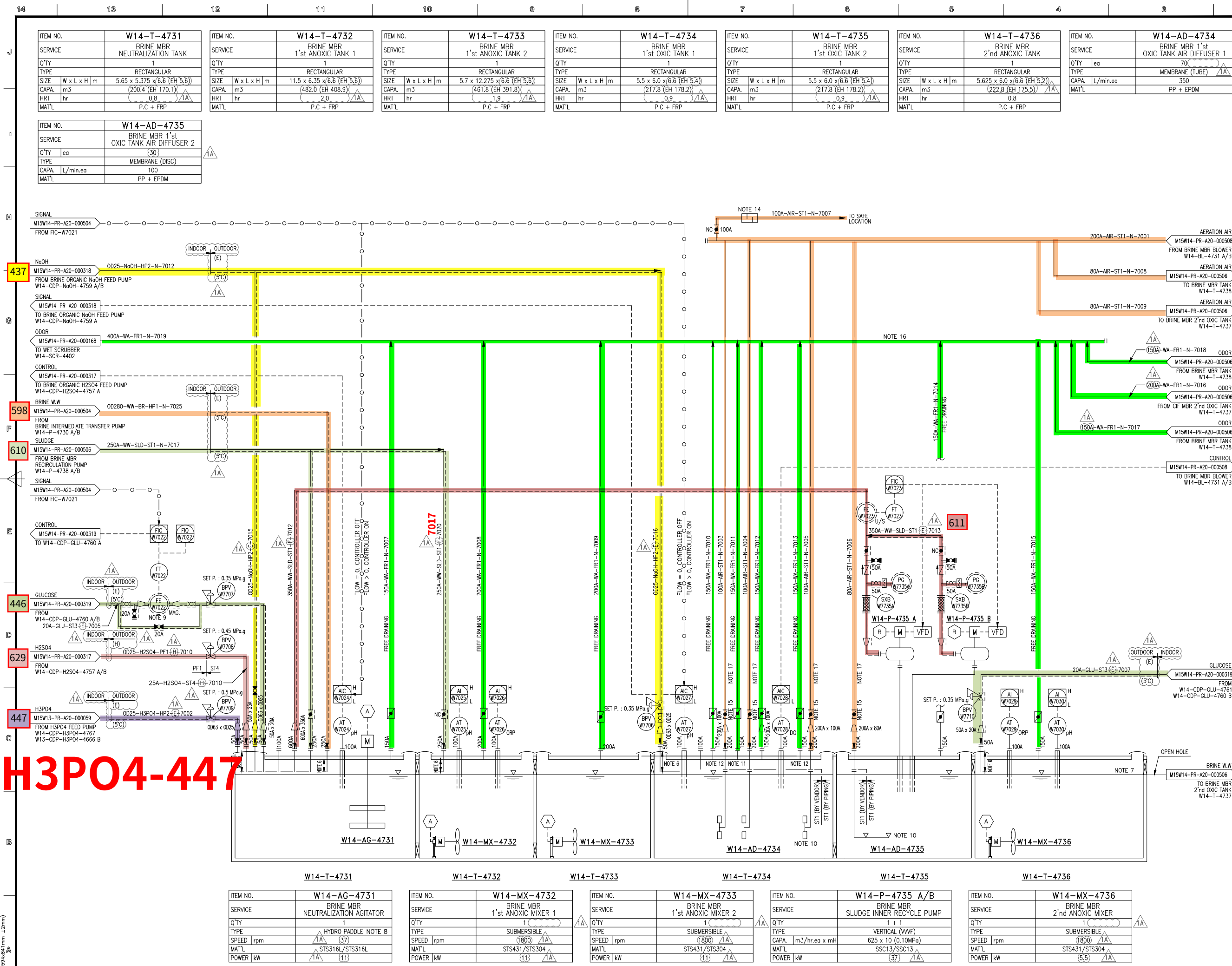
제 도	한 진 희
DRAWN BY	
검 토	심 문 용
CHECKED BY	
승 인	박 동 영
APPROVED BY	
제 작 사	

SK 에코플랜트	
공 사 명	
PROJECT TITLE	
M15X Ph3 WWT Project	
PROJECT NO.	
22330D	

도면명	DWG. TITLE
PIPING & INSTRUMENT DIAGRAM	
H3PO4 STORAGE TANK & FEED PUMP	

속 척	NONE
SCALE	
날 짜	2024. 06. 21
DATE	

PROJECT NO.	22330D	REV.	
도면번호	CJ-M15W13-PR-A20-000059		
SHEET NO.			



대외비

KEY PLAN (1MF)

NOTE

1. TANK의 H/W 이하로 연결되는 모든 배관은 BLOCK 별본 설치될 것.

2. 2m 이상 높은 곳에 설치된 배관은 CHAIN VALVE 또는 PLATFORM 구성할 것.

3. MANHOLE SIZE : 1000W x 1000L x 2EA/TANK MANHOLE COVER : FRP

4. 온전 시 CLOSE 되는 배관에는 LOTO 설치.

5. AGITATOR MAT'L 표기 : SHAFT/BLADE

6. TANK 내 유입시, 유입 확인이 가능하고, 배관이 되지 않도록 최소 길이로 반영 할 것. (SLAB로부터 100mm 떨어질 것)

7. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.

8. 추후 VENDOR 에서 결정한다.

9. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 확정 예정.

10. CIF 1st/2nd OXIC TANK는 선취류 표기가 되도록 배설 해야하나, 건물외곽에서 다른 공간에 설치공간 부족으로 M15X PH4 WWT에는 적용 안함.

11. 1차 호기조 내 AIR 확인이 가능하도록 중간지점에 점검구 반영할 것.

12. 1차 호기조 내 pH, DO 계측기의 설치 위치는 중간 지점보다 앞쪽에 설치할 것.

13. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER

14. LOOSE SUPPLY INSTALLATION BY PURCHASER.

15. AIR용 조정을 위해 GEAR OPERATED BUTTERFLY VALVE를 적용한다.

16. 말뚝이 배관 구성 및 배관 NUMBER는 도면번호 W14-PR-A20-000168을 기준으로 한다.

17. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치할 것. (TAG NO. 아래 참조)

446

REVISED AFC

25.08.14

446

REVISED AFC

25.01.17

446

APPROVAL FOR CONSTRUCTION

24.10.31

수정

REV.

실명

DESCRIPTION

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DATE

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CHECKED BY

한진회

승인

APPROVED BY

박동명

제

작

사

SK 에코플랜트

공사명

PROJECT TITLE

M15X Ph4 WWT Project

도면명

DWG. TITLE

PIPING & INSTRUMENT DIAGRAM
BRINE MBR ANOXIC & OXIC TANK

속

척

SCALE

NONE

날

짜

DATE

2025. 08. 14

PROJECT NO.

22330D

REV.

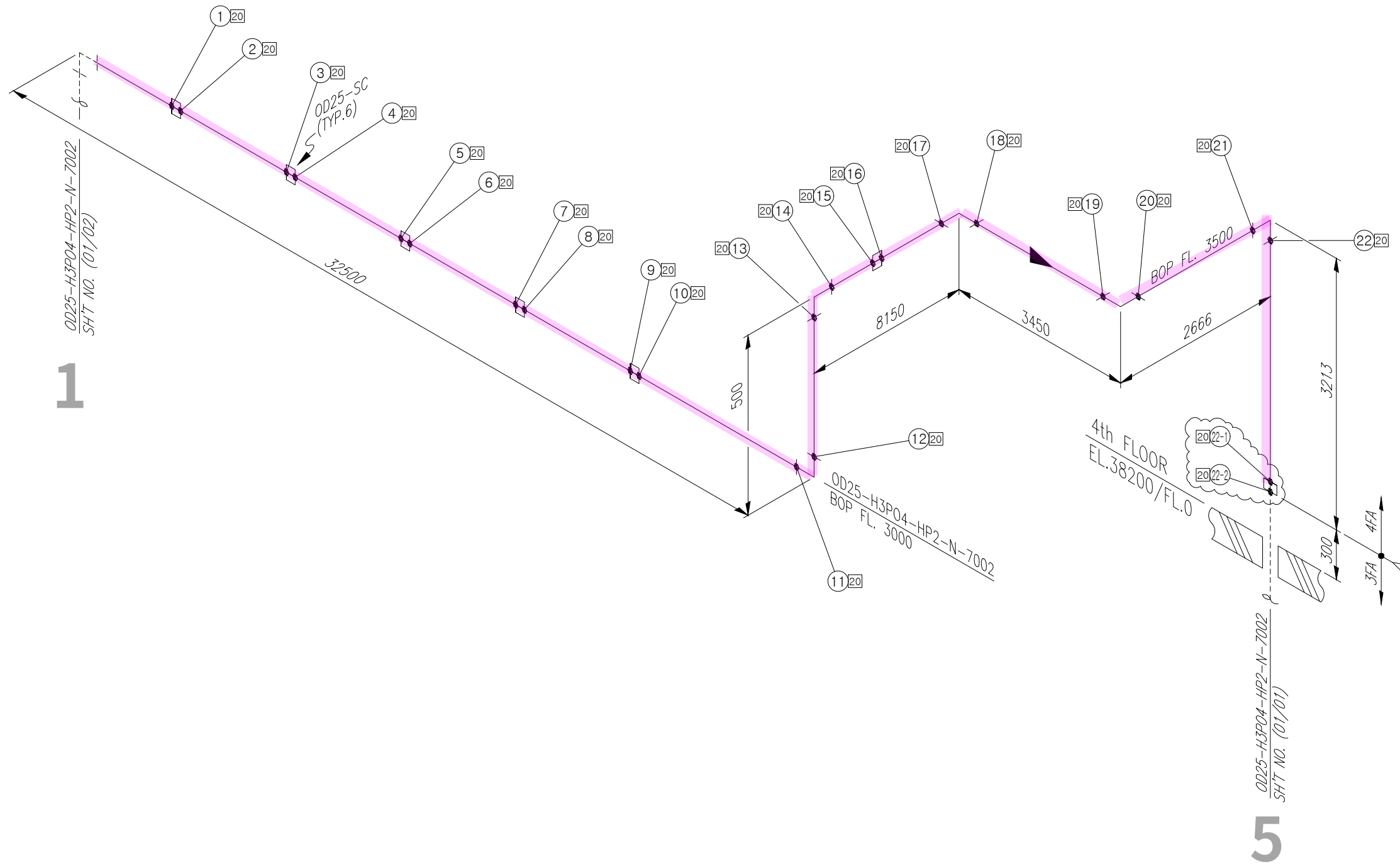
A

도면번호

SHEET NO.

CJ-M15W14-PR-A20-000505

1A

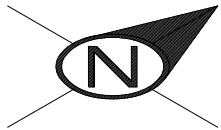


<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

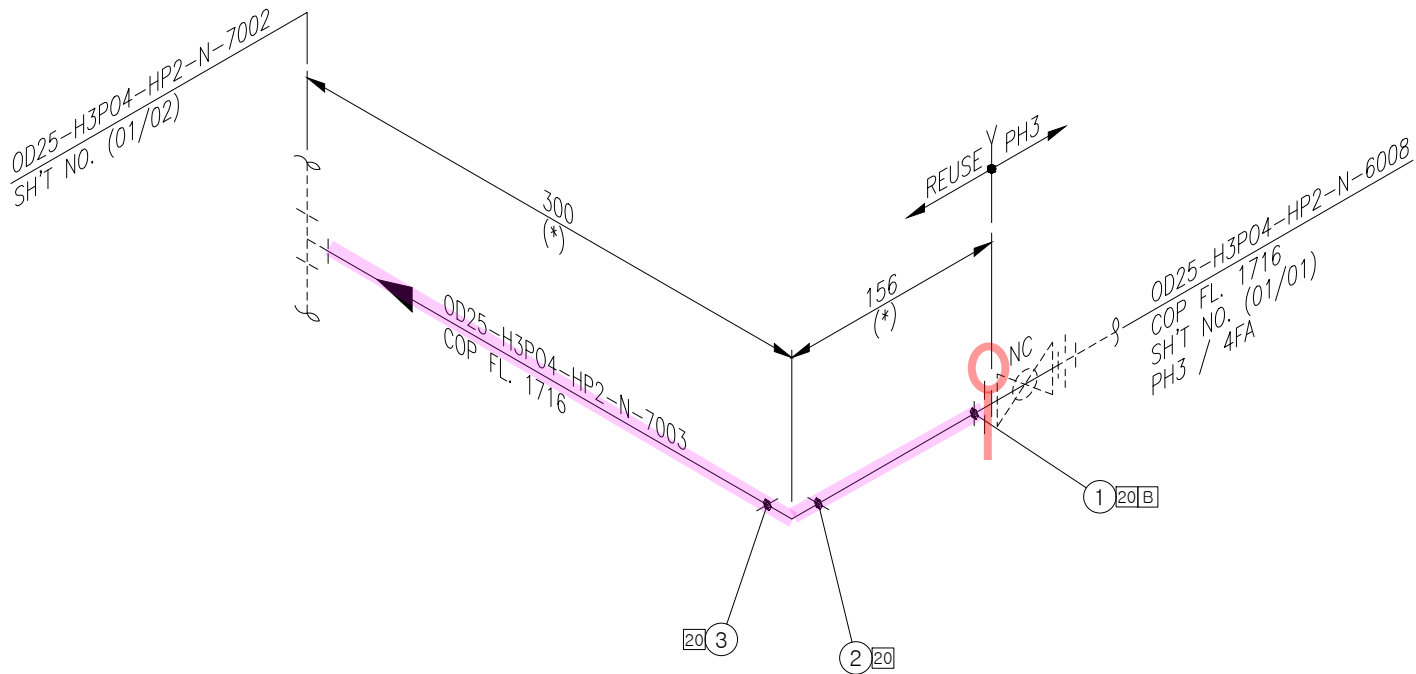
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* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/G1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME						
	THICK.	—	DESIGN	3.3	60	UNIT 4FA								M15X Ph4 Project						
			OPERATING	3.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040007-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	H3P04	JOB NO.					
																			22330D	
						REF. P & ID CJ-M15W14-PR-A20-000159							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	0D25-H3P04-HP2-N-7002



1



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLQ/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1



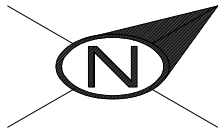
ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD25	0.5M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
E90	OD25	1EA	ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
G	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD25	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SH	OD25	1EA	SHIELD

CLIENT

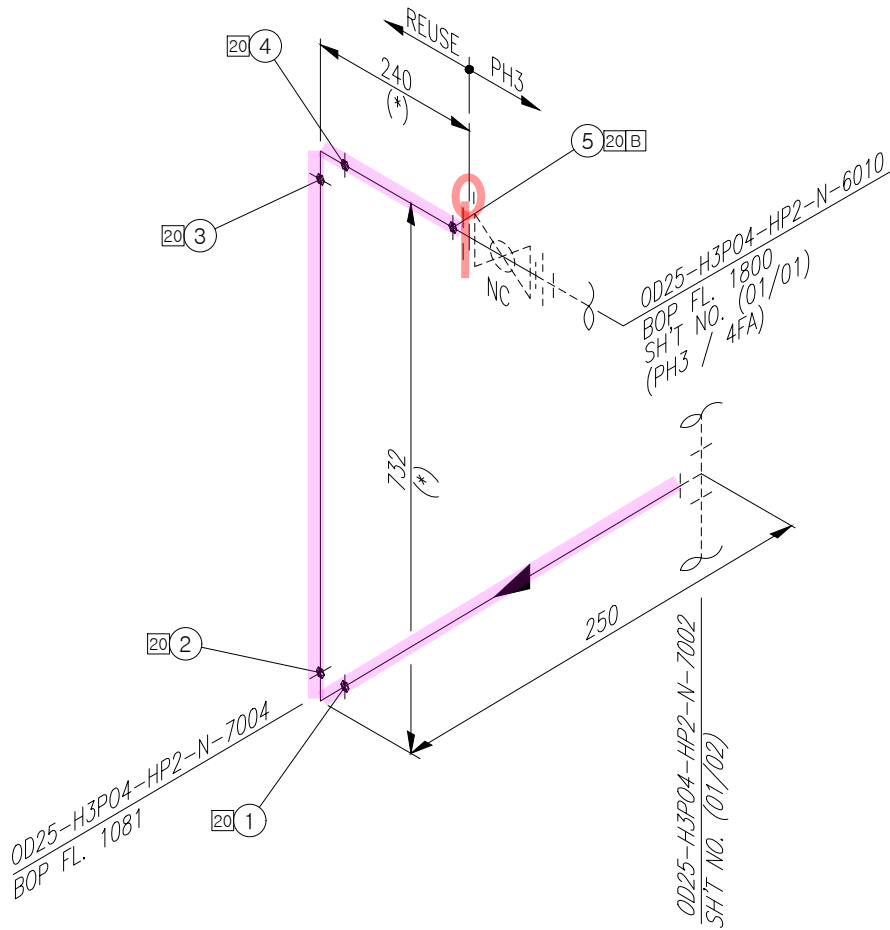


3

CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
H3P04	22330D
LINE NO.	SH'T NO.
OD25-H3P04-HP2-N-7003	01 / 01
	REV.
	0



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD25	1.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
E90	OD25	2EA	ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
G	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD25	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SH	OD25	1EA	SHIELD

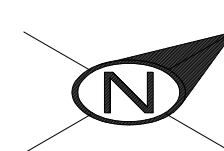


1

- <NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14										CONTRACTOR	PROJECT NAME
	THICK.	—	DESIGN	1.0	60	UNIT	4FA											M15X Ph4 Project
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040009-A											COMMODITY	JOB NO.
									25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영					
						REF. P & ID CJ-M15W13-PR-A20-000059		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				LINE NO.	SH'T NO.
																	OD25-H3P04-HP2-N-7004	01 / 01
																		REV.
																		0



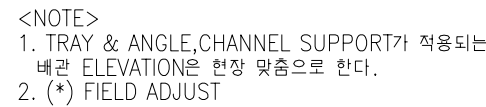
6



5

TOR  SK ecoplant

정 : SK HYNIX의 기밀정보를 자료로서, 어떠한 중도에 동종 업계 외 사용, 복제, 제3자에게 공개, 누설 또는 유출한 경우 수 없으며, 회사나 그 포자의 사명, 평판, 이익을 위해 발생하여 중도에 동종 포자에게 미/할 상채 의무가 없는 경우 동종 포자에게



CLIENT  6

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	60	UNIT 2FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-020011-H							COMMODITY	JOB NO.		
						REF. P & ID CJ-M15W14-PR-A20-000505		24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	0D25-H3P04-HP2-N-7002	SH'T NO.	REV.
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			01 / 01	0

고 : 그 전에도 이미 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 묵적의 사용 통제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 당사는 오폐수의 폐기 사용으로 피해가 발생할 경우 오폐수자에게 민/형사상 책임을 요구할 수 있는

